GRACEFUL LABELINGS OF FIVE-LEAF TREES WITH MULTIPLE BRANCH VERTICES

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Abstract. We prove that every finite tree with exactly five leaves and at least two branch vertices is graceful. Equivalently, we treat the two degree configurations $(4, 3)$ and $(3, 3, 3)$, with all remaining degrees at most two. The proof combines rooted path labelings, amalgamation, symbolic constructions for three-legged spiders, and explicit finite certificates. The central rooted result assigns zero to any internal vertex on a specified leg of a three-legged spider, apart from one excluded configuration. A stronger α-labeling result handles the case in which all four outer pendant paths in the three-branch tree have length two. Separate Lean developments verify the two exact catalog targets, including the structural reductions, infinite parameter ranges, and finite cases.

1. Introduction and main results

A graceful labeling of a tree with q edges assigns distinct integers from 0 to q to its vertices so that the absolute differences across its edges are precisely $1, \dots, q$. The Graceful Tree Conjecture asks whether every finite tree admits such a labeling. A *branch vertex* is a vertex of degree at least three. Our main result treats the two possible configurations of a five-leaf tree with more than one branch vertex.

Theorem 1.1. *Every finite tree with exactly five leaves and at least two branch vertices is graceful.*

For any nontrivial finite tree, the degree-sum identity gives

$$\#\{v : \deg(v) = 1\} = 2 + \sum_{\deg(v)\geq 3} (\deg(v) - 2).$$

If the left side is five and there is more than one branch vertex, the positive summands on the right partition three as $2 + 1$ or $1 + 1 + 1$. Thus the branch degrees are exactly $(4, 3)$ or $(3, 3, 3)$. We prove these two configurations separately, beginning with the former.

Theorem 1.2. *Let T be a finite tree. Suppose that distinct vertices A, B satisfy*

$$\deg_T(A) = 4, \qquad \deg_T(B) = 3, \qquad \deg_T(v) \leq 2 \quad (v \notin \{A, B\}).$$

Then T is graceful.

The degree configurations in Theorem 1.1 are among the five-leaf families discussed by Panpa, Imnang, and Wasuanankul [6]. Rooted labelings and amalgamation provide a common framework for both cases. Theorem 1.2 treats the degree-$(4, 3)$ configuration; Theorem 8.1 treats three degree-three vertices. These correspond to `Math15Catalog.source10` and `Math15Catalog.source11`, respectively. The case of a single degree-five branch vertex is outside the hypotheses of Theorem 1.1.

Let $T(a, b, c; d; e, f)$ denote the tree in Figure 1, where each parameter is a positive integer and measures a path length in edges. Its edge count is

$$q = a + b + c + d + e + f.$$

The path joining A to B has length d; three pendant paths of lengths a, b, c meet A, and two of lengths e, f meet B.

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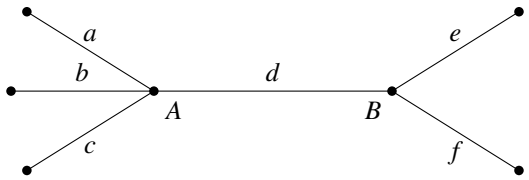


FIGURE 1. The six-parameter tree $T(a, b, c; d; e, f)$. Each segment represents a path of the indicated positive length.

Lemma 1.3 (Structural reduction)**.** *Every tree satisfying the hypotheses of Theorem 1.2 is isomorphic to some $T(a, b, c; d; e, f)$ with all six parameters positive.*

Proof. The tree has a unique simple A–B path. Every internal vertex of this path has degree two, since neither branch vertex occurs internally. Of the edges incident to A, exactly one lies on this path, leaving three components attached at A; similarly two remain at B. Starting along any such edge, there is no further branch vertex, so the component is a path ending at a leaf. Finiteness guarantees that the path ends, and acyclicity prevents it from returning to an earlier vertex. Distinct paths are disjoint outside their attachment vertices, again by acyclicity. Connectivity ensures that these paths contain every vertex. Their lengths give the required parameters. □

We first establish the rooted labeling operations and settle the degree-$(4, 3)$ case. We then strengthen the spider constructions and combine them to treat the degree-$(3, 3, 3)$ case. Section 9 identifies both formal statements and the accompanying checked Lean files.

2. Rooted labelings and elementary operations

Write $[u, v] = \{u, u + 1, \ldots, v\}$ for an integer interval, empty when $u > v$. For a tree with $q \geq 1$ edges, an *α-labeling* is a graceful labeling g with an integer cut $0 \leq k < q$ such that every edge joins a vertex labeled at most k to one labeled greater than k. The two label sets are called lower and upper, respectively. Throughout, P_N denotes a path on N vertices.

We use two established path facts. A prescribed vertex of a nontrivial path can receive zero in an α-labeling, with the sole exception of the middle vertex of P_5; see Rosa [7] and the restatement in [8, Lemma 2(b)]. Also, a prescribed endpoint of P_N can receive any value in $[0, N - 1]$ in a graceful labeling [1]. Every three-legged spider has a graceful labeling with its center zero [6, Theorem 3.2]. These are mathematical inputs to the exposition. Their needed versions are proved within the Lean development, rather than assumed as additional axioms.

If g is an α-labeling of a q-edge tree with cut k, define

$$R(g)(v) = \begin{cases} k - g(v), & g(v) \leq k, \\ q + k + 1 - g(v), & g(v) > k. \end{cases} \tag{1}$$

The two label intervals are permuted separately, and an edge difference j becomes $q + 1 - j$. Thus $R(g)$ is again an α-labeling with cut k. It interchanges the labels 0 and k. A second useful transformation is

$$Z(g)(v) = \begin{cases} q - k + g(v), & g(v) \leq k, \\ g(v) - k - 1, & g(v) > k. \end{cases} \tag{2}$$

It has cut $q - k - 1$, sends label $k + 1$ to zero, and again changes difference j to $q + 1 - j$.

Lemma 2.1 (Amalgamation at zero)**.** *Let G, H be trees with distinguished vertices u, v. Suppose that G has an α-labeling with u labeled zero, and H has a graceful labeling with v labeled zero. Identifying u and v produces a graceful tree.*

Proof. Let $r = |E(G)|$, $s = |E(H)|$, and let k be the cut on G. Apply R to G, so that u has label k. Add s to its upper labels, and add k to every label on H. The vertex intervals are $[0, k]$, $[k, k + s]$, and $[k + s + 1, r + s]$. Their only overlap is the common vertex at k. The differences from H are $[1, s]$, and those from G are $[s + 1, s + r]$. □

This is the required special case of the Huang–Kotzig–Rosa construction [3]; see also [8, Lemma 1].

Lemma 2.2 (A chosen leaf of a four-legged spider)**.** *A four-legged spider has a graceful labeling with zero at any specified leaf.*

Proof. Delete the leg ending at that leaf. Label the remaining three-legged spider with its center zero. Restore the deleted leg one edge at a time. If the current m-edge tree is labeled by h with its current attachment vertex zero, replace every old label by $m + 1 - h$ and label the new vertex zero. The old differences remain $[1, m]$, while the new difference is $m + 1$. The new endpoint is zero, so the operation iterates. □

Lemma 2.3 (Appending a tail)**.** *Let a q-edge tree have an α-labeling g of cut k, and let an attachment vertex have label b. Append a path of $m \geq 1$ new vertices there. A graceful labeling preserving every old lower label exists whenever*

$$m \geq \begin{cases} k + 1 - b, & b \leq k, \\ b - k, & b > k. \end{cases} \tag{3}$$

In particular, $m \geq \max(k + 1, q - k)$ is sufficient for every attachment vertex.

Proof. Add m to all old upper labels and leave the lower labels unchanged. The unused interval is $[k + 1, k + m]$. If $b \leq k$, label the new m-vertex path gracefully on this interval with its first label $b + m$; if $b > k$, choose first label b. The endpoint theorem and (3) permit these choices. The old differences become $[m + 1, m + q]$, the internal path differences are $[1, m - 1]$, and the joining difference is m. The single-vertex case has the same direct interpretation. □

3. A prescribed zero in $S(2, 2, n)$

Let $S(2, 2, L)$ have center B, short legs B, x_1, y_1 and B, x_2, y_2, and long leg $B, w_1, \ldots, w_L$. It has $q = L + 4$ edges. Every certificate below uses the vertex order

$$(B, x_1, y_1, x_2, y_2, w_1, \ldots, w_L). \tag{4}$$

Theorem 3.1 (Prescribed zero on the long leg)**.** *For all positive integers d, c, the spider $S(2, 2, d + c)$ has a graceful labeling in which w_d receives zero.*

We call an α-labeling *canonical* if, with $L = d + c$,

$$(g(B), g(x_1), g(y_1), g(x_2), g(y_2), g(w_1)) = (1, q, 0, q - 1, 2, q - 3). \tag{5}$$

Its cut is $\kappa = 2 + \lfloor L/2 \rfloor$. We seek canonical labelings satisfying

$$g(w_d) = \kappa + (d \bmod 2). \tag{6}$$

When d is even, R sends this label to zero; when d is odd, Z does so. We refer to this step as *normalizing the distinguished vertex*.

3.1. **A symbolic path construction.**

The construction below uses the path-forest viewpoint underlying the finite-state study of bipartite graceful permutations [5]. We give the transitions and parameter choices needed here explicitly.

Proposition 3.2. *If $d, c \geq 1$ and $c \leq 2d - 16$, there is an α-labeled path $(p_0, \ldots, p_{L-1})$ on $[0, L-1]$, $L = d + c$, with $p_0 = 1$. At position $d - 1$ it has the largest lower label when d is odd, and the smallest upper label when d is even.*

Use formal symbols L_i, H_j. At the end, for a path on N vertices, interpret them as i and $N - 1 - j$, respectively. An edge L_iH_j then has difference $N - 1 - (i + j)$. We add edges in increasing order of the index sum, keeping a forest of one or two paths. The main path starts at L_1, and this endpoint is never used by a transition. Let A, B be the largest low and high indices used. A state records the other endpoint of the main path, the endpoints of an extra path when present, and any missing symbol below the two maxima:

State	Main endpoint	Extra endpoints	Missing symbol
TL	L_{A-1}	none	none
TH	H_{B-1}	none	none
UH	H_B	none	H_{B-1}
UL	L_A	none	L_{A-1}
VH	H_B	L_A, H_{B-1}	none
VL	L_A	L_{A-1}, H_B	none
WL	H_B	L_{A-1}, L_A	none
WH	L_A	H_{B-1}, H_B	none
XH	H_{B-1}	L_A, H_B	none
XL	L_{A-1}	L_A, H_B	none

In a terminal state TL or TH the largest index sum already used is $A+B$; in every other state it is $A+B-1$. The following table gives all required transitions. The indices in the added edge refer to the state *before* the transition.

From	To	Added edge	From	To	Added edge
TL	UH	$L_{A-1}H_{B+2}$	TH	UL	$L_{A+2}H_{B-1}$
UH	VH	$L_{A+1}H_{B-1}$	UL	VL	$L_{A-1}H_{B+1}$
VH	TH	L_AH_B	VL	TL	L_AH_B
VH	WL	$L_{A+1}H_{B-1}$	VL	WH	$L_{A-1}H_{B+1}$
WL	TL	L_AH_B	WH	TH	L_AH_B
WL	XH	$L_{A-1}H_{B+1}$	WH	XL	$L_{A+1}H_{B-1}$
XH	VL	$L_{A+1}H_{B-1}$	XL	VH	$L_{A-1}H_{B+1}$

Each transition either extends a path endpoint to a new symbol, creates the extra two-vertex path, or joins the two components. The endpoint table shows that no cycle or vertex of degree three is created. Its index sum is exactly the next unused sum. For example, VH $\to$ WL extends the extra path at H_{B-1} to the new symbol L_{A+1}; the new sum is $A + B$. Updating A gives the stated WL endpoints. The other rows are verified by the same substitution.

The following terminal seeds have the complete index-sum inventories $[0, N_0 - 2]$:

N_0	State	Symbolic path
3	TL	L_1, H_0, L_0
4	TH	L_1, H_1, L_0, H_0
6	TH	$L_1, H_0, L_0, H_2, L_2, H_1$
9	TL	$L_1, H_0, L_0, H_2, L_2, H_1, L_4, H_3, L_3$

The four-step cycles

$$\mathsf{TL} \to \mathsf{UH} \to \mathsf{VH} \to \mathsf{WL} \to \mathsf{TL}, \qquad \mathsf{TH} \to \mathsf{UL} \to \mathsf{VL} \to \mathsf{WH} \to \mathsf{TH}$$

add four vertices. Thus a TL path exists for every odd $N_0 \geq 7$, and a TH path for every even $N_0 \geq 6$.

Enter VH from TL via UH, or enter VL from TH via UL. At entry the main and extra paths have $N_0 + 1$ and 2 vertices. Each of the six-step cycles

$$\mathsf{VH} \to \mathsf{WL} \to \mathsf{XH} \to \mathsf{VL} \to \mathsf{WH} \to \mathsf{XL} \to \mathsf{VH},$$
$$\mathsf{VL} \to \mathsf{WH} \to \mathsf{XL} \to \mathsf{VH} \to \mathsf{WL} \to \mathsf{XH} \to \mathsf{VL}$$

adds two vertices to the main path and four to the extra path. After t cycles their sizes are therefore

$$M = N_0 + 1 + 2t, \qquad E = 2 + 4t. \tag{7}$$

On exiting VH directly to TH, or via WL to TL, the final symbols L_A, H_B occur at zero-based positions $M, M-1$. On exiting VL directly to TL, or via WH to TH, they occur at $M-1, M$. The longer exit VH $\to$ WL $\to$ XH $\to$ VL $\to$ TL puts them at $M, M+1$; its reflected exit from VL puts them at $M+1, M$.

Proof of Proposition 3.2. Write $c = s+4t$, where $1 \le s \le 4$. Choose the following initial size, terminal state, and exit, after entering the appropriate two-path state and completing t cycles:

d	s	N_0	Initial state	Exit destinations
odd	1	$d-2-2t$	TL	TH
odd	2	$d-1-2t$	TH	TL
odd	3	$d-1-2t$	TH	WH, TH
odd	4	$d-2-2t$	TL	WL, XH, VL, TL
even	1	$d-2-2t$	TH	TL
even	2	$d-2-2t$	TH	WH, TH
even	3	$d-1-2t$	TL	WL, TL
even	4	$d-2-2t$	TH	WH, XL, VH, TH

The bound $c \le 2d-16$ ensures that odd initial sizes are at least seven and even initial sizes at least six. Hence all seeds exist. Formula (7) and the exit positions give $d+c$ final vertices and the required boundary label at $d-1$ in each of the eight rows. In the first row, for instance, the final size is $N_0+3+6t = d+c$, and L_A occurs at $N_0+1+2t = d-1$.

At termination all low indices $0, \dots, A$ and high indices $0, \dots, B$ occur, and $A+B = N-2$. Thus the numerical labels are the disjoint intervals $[0, A]$ and $[A+1, N-1]$. Every edge crosses the cut A; the complete index-sum inventory yields differences $[1, N-1]$. The initial symbol L_1 gives $p_0 = 1$. □

3.2. **Converting the path to a spider.** For the path of Proposition 3.2, set

$$g(B) = 1, \quad (g(x_1), g(y_1)) = (q, 0), \quad (g(x_2), g(y_2)) = (q-1, 2), \quad g(w_i) = L + 2 - p_{i-1}. \tag{8}$$

The long-leg labels are $[3, q-2]$. Its internal differences are $[1, L-1]$, and Bw_1 has difference L because $p_0 = 1$. The four short-leg differences are $q, q-1, q-2, q-3$. These inventories verify gracefulness.

The path starts in its lower class, so it has $\lceil L/2 \rceil$ lower and $\lfloor L/2 \rfloor$ upper labels. The map $p \mapsto L+2-p$ exchanges their order. Adding the three lower gadget labels 0, 1, 2 gives cut $\kappa = 2+\lfloor L/2 \rfloor$. The boundary labels specified in Proposition 3.2 become exactly (6). Consequently normalization proves Theorem 3.1 throughout $c \le 2d-16$.

3.3. **Extending a canonical labeling.**

Lemma 3.3 (Four-edge prefix)**.** *A canonical labeling satisfying* (6) *for* (d, c) *gives one for* $(d+4, c)$.

Proof. Keep the new center at 1, give the short legs labels $(q+4, 0)$ and $(q+3, 2)$, and use the long-leg sequence

$$q+1,\ 4,\ q+2,\ 3,\ g(w_1)+2, \dots, g(w_L)+2. \tag{9}$$

The new short-leg differences are $q+1, \dots, q+4$. The four initial long-leg differences are $q, q-3, q-2, q-1$; the next is $q-4$, the former center-to-leg difference. All subsequent differences are unchanged.

The label intervals likewise exhaust $[0, q+4]$. The cut increases by two, as do all shifted old long-leg labels. The new distinguished position is $d+4$ and still satisfies (6). □

The four canonical seeds in Appendix A have $(d,c) = (8,1),(9,1),(10,1),(11,1)$. Lemma 3.3 gives a canonical one-vertex tail for every $d \geq 8$. After normalization, its edge count is $q_0 = d+5$, and its cut is

$$k = \begin{cases} d/2+2, & d \text{ even}, \\ (d+3)/2, & d \text{ odd}. \end{cases}$$

Hence $\max(k+1, q_0-k) \leq \lfloor d/2 \rfloor + 4$. Appending $c-1$ vertices with Lemma 2.3 proves the prescribed-zero result in the second region

$$d \geq 8, \qquad c \geq \lfloor d/2 \rfloor + 5. \tag{10}$$

For $d \geq 13$ we have $\lfloor d/2 \rfloor + 5 \leq 2d-15$, so the two regions cover every positive c.

For $1 \leq d \leq 7$, Appendix A gives zero-pinned α-labelings with initial tail length $c_0 = 2$ for $d = 1$ and $c_0 = 1$ otherwise. The sufficient final-tail thresholds $c_0 + \max(k+1, q_0-k)$ are

$$(6,5,6,6,7,7,8),$$

in order of d. The cases left over are exactly those in Table 1. For each pair, Appendix B supplies

d	Remaining c	d	Remaining c	d	Remaining c
1	1–5	5	1–6	9	3–8
2	1–4	6	1–6	10	5–9
3	1–5	7	1–7	11	7–9
4	1–5	8	1–8	12	9–10

TABLE 1. The sixty-two residual pairs.

a complete labeling tuple. Its entries are a permutation of $[0, d+c+4]$, its edge differences are a permutation of $[1, d+c+4]$, and its entry at index $4+d$ is zero. These finite checks finish the proof of Theorem 3.1.

4. The uniform length-two family

It remains useful to have a direct formula for $T(2,2,2;d;2,2)$. This avoids any additional theorem about general matching expansions or subdivisions in the final reduction.

Number the joining path $0,1,\ldots,d$, with $A=0$ and $B=d$. The three legs at A are $(0,d+1,d+2)$, $(0,d+3,d+4)$, and $(0,d+5,d+6)$; the two at B are $(d,d+7,d+8)$ and $(d,d+9,d+10)$. Put

$$\varepsilon = d \bmod 2, \qquad Q = d+11-\varepsilon, \qquad r = 2\lfloor d/4 \rfloor + 2.$$

For $0 \leq j \leq d$, define

$$F(j) = \begin{cases} j/2, & j \equiv 0 \pmod 4, \\ d+(4+6\varepsilon)-\lfloor j/2 \rfloor, & j \equiv 1 \pmod 4, \\ j/2+6, & j \equiv 2 \pmod 4, \\ d+(10-6\varepsilon)-\lfloor j/2 \rfloor, & j \equiv 3 \pmod 4. \end{cases} \tag{11}$$

At the three legs of A, assign

$$\big(F(d+1),\ldots,F(d+6)\big) = (Q-1,1,Q-3,3,Q-5,5). \tag{12}$$

If $d \equiv 0$ or $3 \pmod 4$, assign

$$\big(F(d+7),\ldots,F(d+10)\big) = (r, Q-r, r+2, Q-r-2); \tag{13}$$

otherwise reverse each of these two pairs.

Proposition 4.1. *For every $d \geq 1$, formulas (11)–(13) define a graceful labeling of $T(2,2,2;d;2,2)$.*

Proof. Write $d = 4h + s$, $0 \leq s < 4$. Splitting the joining-path indices by their residues modulo four in (11) gives four step-two intervals. Together with (12) and (13), these form a disjoint partition of $[0, d + 10]$. For a direct verification of the edge inventory, define $\Delta(j) = |F(j) - F(j-1)|$ on the joining path. Substitution gives

$$\Delta(j) = \begin{cases} d + 5 - j, & j \not\equiv d \pmod 2, \\ d + 11 - j, & j \equiv d \pmod 2,\ j \bmod 4 < 2, \\ |d - 1 - j|, & j \equiv d \pmod 2,\ j \bmod 4 \geq 2. \end{cases} \tag{14}$$

These values are distinct. The values missing from $[1, d + 10]$ are precisely

s	Differences missing from the joining path
$0, 1$	$\{2,3,4,7\} \cup \{4h+1, 4h+5, 4h+6, 4h+8, 4h+9, 4h+10\}$
$2, 3$	$\{2,4,5,9\} \cup \{4h+3, 4h+7, 4h+8, 4h+10, 4h+11, 4h+12\}$.

Indeed, these identities follow by writing $j = 4i + t$ in (14); each resulting sequence has step four, with the final absolute value contributing 1 when $s = 2$ or 3. The six differences on the legs at A are

$$Q - 1, Q - 2, Q - 3, Q - 6, Q - 5, Q - 10.$$

The four differences on the legs at B are $(2, 7, 4, 3)$ for $s = 0, 1$ and $(4, 9, 2, 5)$ for $s = 2, 3$. They fill exactly the displayed gaps.

The vertex inventory can be checked in the same way. The labels missing from the joining path in $[0, d + 10]$ are

s	Labels missing from the joining path
$0, 1$	$\{1, 3, 5, 2h+2, 2h+4, 2h+7, 2h+9, 4h+6, 4h+8, 4h+10\}$
$2, 3$	$\{1, 3, 5, 2h+2, 2h+4, 2h+9, 2h+11, 4h+8, 4h+10, 4h+12\}$.

Formulas (12) and (13) supply these ten labels exactly once. Thus all vertex labels and all edge differences have the required inventories. □

5. Completion of the degree-$(4,3)$ case

Proof of Theorem 1.2. By Lemma 1.3, write the tree as $T(a, b, c; d; e, f)$.

If $(e, f) \neq (2, 2)$, the two legs at B form a path with an α-labeling having $B = 0$: the sole excluded zero position would require $e = f = 2$. The other component, keeping B as a leaf, is the four-legged spider with lengths a, b, c, d. Lemma 2.2 gives it a graceful labeling with $B = 0$. Lemma 2.1 now labels their union gracefully.

Suppose $e = f = 2$ and $(a, b, c) \neq (2, 2, 2)$. Permute the three legs at A so that $(a, b) \neq (2, 2)$. Their union is a path with an α-labeling having $A = 0$. The remaining component is $S(2, 2, d + c)$ centered at B, with $A = w_d$ on its long leg. Theorem 3.1 supplies a graceful labeling with $A = 0$. Another application of Lemma 2.1 completes this case.

Finally, when $a = b = c = e = f = 2$, Proposition 4.1 applies. The three cases exhaust all positive parameter choices. □

6. Alpha labelings with a prescribed zero in $S(2,2,n)$

For the second branching pattern we need a stronger form of Theorem 3.1: the labeling must sometimes be an α-labeling. We retain the spider notation and vertex order (4).

Lemma 6.1 (Prescribed lower endpoint)**.** *Let N, s be nonnegative integers with $N \geq 2s+1$ and $N \neq 4s+1$. There is an α-labeling $(p_0, \ldots, p_{N-1})$ of P_N, with cut $\lfloor (N-1)/2 \rfloor$ and $p_0 = s$. Its other endpoint can be chosen as*

$$p_{N-1} = \begin{cases} N/2 + s, & N \text{ even}, \\ \lfloor N/2 \rfloor - s, & N \text{ odd}. \end{cases}$$

Proof. For $N = 2h$, this is the prescribed-endpoint α-path theorem of Cattell [1]. Here $0 \leq s < h$, the cut is $h-1$, and the endpoints are $s, h+s$. We describe the odd case to make its exceptional equality explicit. Write $N = 2h+1$. Suppose first that $2s > h$. Take the even α-path on $2s$ vertices whose endpoints are $2s-h-1$ and $3s-h-1$. Add $h-s+1$ to all its labels and append the alternating sequence

$$0,\ 2h,\ 1,\ 2h-1,\ \ldots,\ h-s.$$

The first block uses $[h-s+1, h+s]$; the appended block uses $[0, h-s] \cup [h+s+1, 2h]$. Their internal differences are, respectively, $[1, 2s-1]$ and $[2s+1, 2h]$, and the joining difference is $2s$. Every edge crosses h, and the endpoints are $s, h-s$. If $2s < h$, carry out this construction with $h-s$ and apply the transformation R of (1), with cut h. The omitted case $2s = h$ is precisely $N = 4s+1$. □

Lemma 6.2 (Even tail preserving alpha)**.** *Suppose a q-edge tree has an α-labeling of cut k, and the attachment vertex has label t. Put*

$$r = \begin{cases} k+1-t, & t \leq k, \\ t-k, & t > k. \end{cases}$$

Appending $2h$ new vertices preserves an α-labeling and every old lower label whenever $h \geq r$. The new cut is $k+h$. In particular, $h \geq \max(k+1, q-k)$ suffices.

Proof. Take an α-labeled P_{2h} with cut $h-1$ and first label $r-1$, using Lemma 6.1. If $t \leq k$, complement its labels, so its first label is $2h-r$. Keep old lower labels, add $2h$ to old upper labels, and add $k+1$ to the path labels. The three label intervals are $[0, k]$, $[k+1, k+2h]$, and $[k+2h+1, q+2h]$. The old differences are shifted to $[2h+1, q+2h]$; new internal differences are $[1, 2h-1]$. The first new label is $t+2h$ when $t \leq k$, and t otherwise, so the joining difference is $2h$. The first new vertex belongs to the opposite side of the cut $k+h$ from the attachment vertex, and all other edges cross that cut as well. □

Theorem 6.3. *Let $d, c \geq 1$. If*

$$(d, c) \notin \{(1,1), (1,4), (1,5)\},$$

then $S(2, 2, d+c)$ has an α-labeling with $w_d = 0$.

Proof. The canonical construction in Proposition 3.2 and (8), followed by R or Z, already proves the stronger α conclusion throughout $c \leq 2d - 16$.

For the other infinite region we use seeds with final tail length $b = 1$ or 2. The four $b = 1$ seeds are in Appendix A. The following are the four $b = 2$ seeds; each row gives a canonical labeling, with its boundary cut K.

d	K	Label tuple in order (4)
8	7	(1, 14, 0, 13, 2, 11, 4, 12, 3, 9, 5, 10, 7, 8, 6)
9	7	(1, 15, 0, 14, 2, 12, 4, 13, 3, 10, 6, 11, 5, 8, 7, 9)
10	8	(1, 16, 0, 15, 2, 13, 5, 10, 6, 12, 3, 14, 4, 11, 8, 9, 7)
11	8	(1, 17, 0, 16, 2, 14, 3, 15, 5, 11, 6, 13, 4, 12, 8, 9, 7, 10)

The four-edge prefix, Lemma 3.3, gives both seed families for every $d \geq 8$. After normalization their cuts are

$$k = \lfloor d/2 \rfloor + b + 1, \qquad \max(k+1, d+b+4-k) \leq \lfloor d/2 \rfloor + 4. \tag{15}$$

For $2 \leq d \leq 7$, the twelve seed cases $b \in \{1, 2\}$ belong to the finite certificate set described below; they have the same cuts and satisfy the same bound.

Now suppose $d \geq 2$ and $c \geq 2\lfloor d/2 \rfloor + 9$. Choose $b = 1$ for odd c, and $b = 2$ for even c. Then $c - b$ is even and

$$c - b \geq 2\big(\lfloor d/2 \rfloor + 4\big).$$

Lemma 6.2 and (15) extend the corresponding seed to tail length c, retaining both the zero and the α property.

For $d = 1$, two zero-pinned seeds, both with cut 3, are

c	label tuple
2	$(5, 3, 4, 2, 6, 0, 7, 1)$
3	$(5, 3, 4, 2, 6, 0, 8, 1, 7)$.

The uniform half-length bounds in Lemma 6.2 are 4 and 5, respectively. Thus these seeds cover every even $c \geq 10$ and every odd $c \geq 13$.

For $d \geq 23$ the first two infinite regions cover every positive c, because $2d - 15 \geq 2\lfloor d/2 \rfloor + 9$. The exact residual set is

$$\begin{aligned} C = {} & \{(1,2), (1,3), (1,6), (1,7), (1,8), (1,9), (1,11)\} \\ & \cup \{(d,c) : 2 \leq d \leq 22, \quad 1 \leq c \leq 2\lfloor d/2 \rfloor + 8, \quad c > 2d - 16\}. \end{aligned} \tag{16}$$

There are 207 pairs. The accompanying `Source11.lean` contains their complete label tuples under the names `h2_d_c_labels`, with the decimal values of d, c substituted into the name. Their certificates are `h2_d_c_certificate`, and the exhaustive finite case split is `h2_finite_region`.

For precision, this finite verification consists solely of the following integer checks. Put $q = d + c + 4$, and set

$$\pi(j) = \begin{cases} 0, & j \in \{1, 3, 5\}, \\ j - 1, & \text{otherwise}, \end{cases} \qquad k = \begin{cases} 2 + \lfloor (d+c)/2 \rfloor, & d \text{ even}, \\ 1 + \lceil (d+c)/2 \rceil, & d \text{ odd}. \end{cases}$$

For each tuple $(\ell_0, \ldots, \ell_q)$ one checks that

$$\begin{aligned} \operatorname{sort}(\ell_0, \ldots, \ell_q) &= (0, \ldots, q), \qquad \ell_{d+4} = 0, \\ \operatorname{sort}\big(|\ell_j - \ell_{\pi(j)}| : 1 \leq j \leq q\big) &= (1, \ldots, q), \\ (\ell_j \leq k) &\iff (\ell_{\pi(j)} > k) \qquad (1 \leq j \leq q). \end{aligned}$$

The tuples satisfy all these checks; their parameter set is exactly (16). This also verifies the small seeds used above and completes the proof. □

7. A prescribed zero in a three-legged spider

We need a rooted version of the three-legged-spider theorem. Write $F(a, b; d, c)$ for $S(a, b, d + c)$, with its distinguished vertex at distance d from the center on the third leg. All four parameters are positive. For labeling certificates we use the following order: first the vertices $x_0, \ldots, x_{a+b}$ of the path formed by the first two legs, with center x_a, then the vertices $y_1, \ldots, y_{d+c}$ of the third leg. The distinguished vertex is y_d.

Theorem 7.1 (Rooted three-legged spiders)**.** *The tree $F(a, b; d, c)$ has a graceful labeling with $y_d = 0$, provided*

$$(a, b, d, c) \neq (1, 1, 1, 2).$$

The exception is genuine. If the first vertex of the long leg of $S(1, 1, 3)$ had label zero, let u, p, q be the center and its two leaf labels, and let v, z be the two subsequent long-leg labels. These are $1, \ldots, 5$, and the differences are $u, |u - p|, |u - q|, v, |v - z|$. Difference five forces $u = 5$ or $v = 5$. If $u = 5$, summing the other differences gives $2v + z + |v - z| = 10$, forcing $(v, z) = (2, 4)$; then a difference is repeated. If $v = 5$ and $u = 4$, the remaining three differences sum to seven rather than six. If $v = 5$ and $u \neq 4$, difference four forces $z = 1$; the only possibilities $u = 2, 3$ again repeat a difference.

Only an ordinary graceful labeling is asserted. The proof uses two explicit path constructions, elementary attachment operations, and finite certificates. We give the constructions and the complete parameter cover; the finite lists and their individual checks are retained in the companion `Source11.lean`. Thus the finite calculations below are part of the proof, not evidence extrapolated to untested parameters.

7.1. **Attaching an alpha path to an ordinary graceful tree.** Suppose a tree with q edges has a graceful labeling f, and a path on m new vertices has an α-labeling $p_0, \dots, p_{m-1}$ with cut k. Attach its first vertex to a vertex z of the tree. Give the old vertices labels $k+1+f$, and give the new vertices labels

$$\widehat{p_i} = \begin{cases} p_i, & p_i \le k, \\ p_i + q + 1, & p_i > k. \end{cases} \tag{17}$$

The old differences are $1, \dots, q$ and the internal new differences are $q+2, \dots, q+m$. The vertex intervals are disjoint and exhaust $[0, q+m]$. Consequently this is graceful if the joining difference is $q+1$, which is ensured by either of the identities

$$\begin{array}{ll} p_0 \le k, & f(z) = q + p_0 - k, \\ p_0 > k, & f(z) = p_0 - k - 1. \end{array} \tag{18}$$

A zero on the new path remains zero.

Apply this with the old tree $x_0 \dots x_q$, $q = a + b$, $z = x_a$, $m = d + c$, and $p_{d-1} = 0$. Counting the two color classes gives

$$k = \left\lfloor \frac{d-1}{2} \right\rfloor + \left\lfloor \frac{c}{2} \right\rfloor, \qquad p_0 \le k \iff d \text{ is odd.} \tag{19}$$

In particular, if an α-path has

$$p_{d-1} = 0, \qquad p_0 = \begin{cases} k - r, & d \text{ odd}, \\ k + 1 + r, & d \text{ even}, \end{cases} \tag{20}$$

then it suffices to give x_a label $q - r$ or r, respectively. The lower case includes the condition $r \le k$.

We use the following small prescribed labels on ordinary paths: every vertex can receive 0; every vertex can receive 1, except the middle of P_3; and every vertex can receive 2, except the middle of P_5 (with the order large enough to contain the specified label). Here the last two statements are needed only when $q > 2$ and $(a, b) \ne (2, 2)$, so neither exception arises. For clarity, these are statements about ordinary graceful paths.

One constructive verification of these facts is useful for reproducibility. Let $W_n(j) = j/2$ for even j and $W_n(j) = n - 1 - \lfloor j/2 \rfloor$ for odd j. To put t at position i, put $P = i + 2t + 1$ and concatenate

$$W_n(P-1), W_n(P-2), \dots, W_n(0),\ \lceil P/2 \rceil + v_0, \dots, \lceil P/2 \rceil + v_{n-P-1}, \tag{21}$$

where v is a graceful path with first label $n - P - \lceil P/2 \rceil$. This is possible when $P < n$ and $\lceil P/2 \rceil \le n - P$. The outer differences are $n - P + 1, \dots, n - 1$, the joining difference is $n - P$, and the inner differences are $1, \dots, n - P - 1$. After reversing positions if necessary, these inequalities hold for every i when $n \ge 12t + 5$. For $t = 1, 2$, the remaining assertions are finite: $t < n < 12t + 5$, $0 \le 2i \le n - 1$, excluding $(n, i, t) = (3, 1, 1), (5, 2, 2)$. They are established in `CattellFixed`, using (21), elementary path symmetries, and 44 listed small path certificates. For $t = 0$, the prescribed-zero alpha-path theorem applies, except at the middle of P_5, where $(1, 4, 0, 2, 3)$ is an ordinary graceful labeling.

7.2. **Two path constructions.** We record the exact auxiliary assertions used in the parameter cover. Their finite data are specified below.

Lemma 7.2 (Paths with a controlled first label)**.** *(1) If $d \geq 10$ and $1 \leq c \leq 2d$, there is an α-path satisfying (20), with $r = 1$ when $c \leq 2d - 16$, and $r = 2$ otherwise. If $1 \leq d \leq 9$, $1 \leq c \leq 2d$, and $(d, c) \neq (3, 2)$, the same assertion holds for some $r \in \{0, 1, 2\}$.*

(2) If $d \geq 2$ and $c > 2d$, there is an α-path with

$$p_{d-1} = 0, \qquad p_0 = \begin{cases} r, & d \text{ odd}, \\ d + c - 1 - r, & d \text{ even}, \end{cases} \tag{22}$$

for each $r = 1, 2$, except that neither value is asserted for $(d, c) = (2, 6)$, and $r = 2$ is not asserted for $(d, c) = (2, 7)$.

Proof. The $r = 1$ part of the first assertion is Proposition 3.2, normalized at the designated boundary label. We describe the additional construction for $r = 2$, using the same symbols L_i, H_j and index sums. An $r = 2$ core consists of two alternating words P, Q, together using each symbol in $L_0, \ldots, L_A$ and $H_0, \ldots, H_B$ once and having index sums $0, \ldots, A + B - 1$ once. The word P starts at L_2. For type VH_2 its last symbol is H_B, and Q runs from L_A to H_{B-2}. For type VL_2 its last symbol is L_A, and Q runs from L_{A-2} to H_B. The direct terminal words are PQ and PQ^{rev}, respectively; their new joining sum is $A + B$.

The following cycles preserve the type, increase both maxima by three, and increase $(|P|, |Q|)$ by $(2, 4)$:

	P'	Q'
VH_2	P, L_{A+1}, H_{B+3}	$L_{A+3}, H_{B+2}, Q, L_{A+2}, H_{B+1}$
VL_2	P, H_{B+1}, L_{A+3}	$L_{A+1}, H_{B+2}, Q, L_{A+2}, H_{B+3}$.

(23)

In each row the six added sums are exactly $A + B, \ldots, A + B + 5$, which proves the invariant directly. For every even $\delta \in [-14, 2]$, one seed of each type has $|Q| - 2|P| = \delta$. Their main-word lengths, in increasing order of δ, are

δ	-14	-12	-10	-8	-6	-4	-2	0	2
VH_2	8	8	8	6	6	6	4	4	4
VL_2	9	9	7	5	7	5	3	5	3.

(24)

These are the 18 finite core certificates `seed_vh_0`, …, `seed_vh_8`, and `seed_vl_0`, …, `seed_vl_8` in `PathTwoConstruction`. If $2d - 15 \leq c \leq 2d$, choose

$$(|P|, |Q|) = \begin{cases} (d, c), & c \text{ even}, \\ (d - 1, c + 1), & c \text{ odd}, \end{cases}$$

using VH_2 when d, c have the same parity and VL_2 otherwise. The required δ lies in the table, and the cycle reaches the target lengths because $d \geq 10$ and the starting main-word lengths are at most nine. The direct exit has its required boundary symbol at position $d - 1$. Normalize by $L_i \mapsto A - i$, $H_j \mapsto A + 1 + j$ when d is odd, and by $L_i \mapsto B + 1 + i$, $H_j \mapsto B - j$ when d is even. This gives (20) with $r = 2$. The remaining short cases are the 89 certificates indexed by $1 \leq d \leq 9$, $1 \leq c \leq 2d$, $(d, c) \neq (3, 2)$ in `SmallShortPaths`. Each checks precisely (20) and the alpha-path inventory.

For the second assertion first suppose $2c \geq d + 24$. The $r = 1$ construction is obtained by applying the first symbolic construction with parameters $(c + 1, d - 1)$, retaining its terminal endpoint, and reversing the word. Its terminal endpoint is one index below a boundary, so normalization gives (22). For $r = 2$ we need an additional prefix operation. Shift every old index by three and prepend

$$L_2, H_2, L_1, H_1, L_0, H_0$$

to P, leaving the shifted Q separate. The five internal prefix sums are $4, 3, 2, 1, 0$, and its joining sum to the shifted initial L_5 is 5; all old sums increase by six. The core type is preserved, and the length change

is $(6, 0)$. Twelve finite seeds suffice: for $|Q| = 2$ and 4 the main-word lengths are

	$\lvert Q\rvert = 2$	$\lvert Q\rvert = 4$
VH_2	$12, 8, 10$	$6, 8, 10$
VL_2	$13, 9, 5$	$7, 3, 11.$

They are `short_seed_0`, …, `short_seed_11` in `PathTwoShort`. For a target even extra length $q_0 \geq 2$, select a seed extra length $e_0 \in \{2, 4\}$ congruent to q_0 modulo four, and put $t = (q_0 - e_0)/4$. Select the seed main length p_0 congruent to $p - 2t$ modulo six. Whenever $p - 2t \geq 13$, apply $(p - 2t - p_0)/6$ prefix operations and t cycles. For the present application take

$$p = c + (d \bmod 2), \qquad q_0 = d - (d \bmod 2),$$

and type VH_2 if $d+c$ is even, VL_2 otherwise. The inequality $2c \geq d+24$ ensures $p-2t \geq 13$. The direct exit has a boundary at position c and its terminal endpoint two indices below a boundary. Normalize at that boundary and reverse the word. This proves (22) with $r = 2$.

Finally, $d \geq 2$, $c > 2d$, $2c < d + 24$ gives exactly 27 pairs:

d	2	3	4	5	6	7
c	5–12	7–13	9–13	11–14	13–14	15.

The 51 certificates in `NeighborFinite` supply both choices of r, except the three omitted choices stated in the lemma. □

7.3. **The parameter cover.** Assume first that $(a, b) \notin \{(1, 1), (2, 2)\}$, and exchange a, b so $a \leq b$. Put

$$q = a + b, \quad U = \lfloor a/2 \rfloor + \lfloor b/2 \rfloor, \quad V = q - U, \quad C(a, b, d) = \begin{cases} U + \lfloor d/2 \rfloor, & d \text{ even}, \\ V + \lfloor d/2 \rfloor, & d \text{ odd}. \end{cases}$$

There are three immediate sufficient conditions.

(1) If $c \leq 2d$, use Lemma 7.2(1), the available path labels 0, 1, 2, and (18). For the excluded path pair $(d, c) = (3, 2)$, the outer construction below works if $a \geq \lfloor b/2 \rfloor + 3$ or $b \geq \lfloor a/2 \rfloor + 3$. If both fail, $1 \leq a \leq b \leq 4$, and the only remaining pairs are $(1, 2), (2, 3), (3, 3), (4, 4)$. Their four complete fork certificates are in `ForkThreeTwo`.

(2) If $c \geq C(a, b, d) + 1$, alpha-label the a, b path with its center zero and cut U. Append the first d edges one at a time, each time complementing the old labels in the new edge count and assigning zero to the new endpoint. This preserves the alpha property; its cut obeys $k_{j+1} = q + j - k_j$, hence $k_d = C(a, b, d)$. The tail-extension lemma, Lemma 2.3, attaches the final c edges at this zero, preserving it.

(3) Define $O(a, b, d, c)$ by

$$(b + d, c) \neq (2, 2), \qquad a \geq \begin{cases} \lfloor (b + d)/2 \rfloor + \lfloor c/2 \rfloor, & d \text{ even}, \\ b + d + c - \lfloor (b + d)/2 \rfloor - \lfloor c/2 \rfloor, & d \text{ odd}. \end{cases} \tag{25}$$

If this holds, alpha-label the path of lengths $b+d, c$ with $y_d = 0$. Its cut is $K = \lfloor (b+d)/2 \rfloor + \lfloor c/2 \rfloor$. The vertex at which the a-leg attaches has a nonzero label, and is lower exactly when d is even. Thus (25) implies the tail-extension bound there: at most K new edges in the lower case, at most $b + d + c - K$ in the upper case. Append the a-leg. The same argument applies with a, b exchanged.

It remains to consider

$$c > 2d, \quad c < C(a, b, d) + 1, \quad \neg O(a, b, d, c), \quad \neg O(b, a, d, c). \tag{26}$$

We give a path splice that covers this entire region. For positive integers ℓ, m and a nonnegative integer u, define $S(\ell, m, u)$ by

$$u \leq \lfloor \ell/2 \rfloor, \quad \ell \neq 4u, \quad 2(\lfloor \ell/2 \rfloor - u) + 1 \leq m, \quad m \neq 4(\lfloor \ell/2 \rfloor - u) + 1. \tag{27}$$

Let $\mathcal{S}^*(a,b,u)$ hold if $\mathcal{S}(a,b,u)$ or $\mathcal{S}(b,a,u)$ holds, or if $u \le U$ and one of $\mathcal{S}(a,b,U-u)$, $\mathcal{S}(b,a,U-u)$ holds. Then $\mathcal{S}^*(a,b,u)$ produces an alpha labeling of $x_0,\dots,x_q$ with $x_a = u$. Indeed, take an alpha path on $\ell+1$ vertices ending in u, and one on m vertices beginning in $m-1-(\lfloor \ell/2 \rfloor - u)$. Lemma 6.1 supplies both under (27). Translate the second path by $\lfloor \ell/2 \rfloor + 1$ and increase the upper labels of the first by m. Their joining difference is m, and their other differences partition $[1, m-1]$ and $[m+1, m+\ell]$. The joined path is alpha with cut U and retains the label u at the join. Reverse its vertex order and/or apply R to obtain the other three choices in $\mathcal{S}^*$.

Here are the exact arithmetic facts used to apply this splice. Under (26) and $1 \le a \le b$:

$$\begin{aligned} &d = 1: \quad \mathcal{S}^*(a,b,\lfloor c/2 \rfloor), \text{ unless } (a,b,c) = (4,4,3) \text{ or } (5,5,6);\\ &d \ge 2: \quad \mathcal{S}^*(a,b,s) \text{ or } \mathcal{S}^*(a,b,s-1), \quad s = \lfloor (d+c-3)/2 \rfloor. \end{aligned} \tag{28}$$

These are integer inequalities, including the displayed exceptional equalities in $\mathcal{S}$. They can be verified by writing each of a, b, c, d as twice an integer plus its parity and expanding the four alternatives in $\mathcal{S}^*$. The complete symbolic checks are the eight parity cases for $d = 1$ and sixteen cases for $d \ge 2$ in `H1SpliceCoverage`. They prove the implications for all positive integer parameters satisfying the stated hypotheses; no bounded numerical sampling is used in (28).

For $d = 1$, take a zero-endpoint alpha path on $c+1$ vertices. Its cut is $\lfloor c/2 \rfloor$; complement the splice labeling and apply (18). For $d \ge 2$, choose the $r = 1$ or $r = 2$ path of Lemma 7.2(2). Equations (19) and (18) require the old center label, up to complementation, to be exactly $s + 1 - r$, which is supplied by (28). For $(d,c) = (2,6)$ or $(2,7)$, (26) forces $a = b = 6$. The latter pair uses the available $r = 1$ path and $\mathcal{S}^*(6,6,3)$; the former is a complete finite certificate. Together with the two $d = 1$ exceptions, the three certificates needed here are precisely

$$(a,b,d,c) = (4,4,1,3),\ (5,5,1,6),\ (6,6,2,6),$$

provided in `NeighborFinite`.

7.4. **The two equal short-arm cases and the finite checks.** If $a = b = 2$, Theorem 3.1 already proves the required ordinary graceful assertion for every $d, c > 0$. We finish with $a = b = 1$. The long-tail construction above remains valid. For the other direction, let $E(b,k)$ mean that $S(1,1,b)$ has a graceful labeling with its third-leg endpoint labeled k; when $b = 0$ this means the center of P_3. Complementation sends $E(b,k)$ to $E(b, b+2-k)$. The alpha-endpoint lemma and (17) give

$$E(b,k) \implies E(b+N,k) \quad \text{if } N \ge 2k+1, \quad N \ne 4k+1. \tag{29}$$

To see the joining identity, reverse the alpha path whose first label is k. Its first label is $N/2 + k$ for even N, and $\lfloor N/2 \rfloor - k$ for odd N. Use the old labeling in the even case and its complement in the odd case in (18). Its final label k remains unchanged.

The zero-endpoint extension gives $E(b,0)$ for all $b \ge 0$. Four small certificates give $E(b,1)$ for $b = 1, 2, 3, 6$:

b	labels in the stated path-then-leg order
1	$(0,3,2,1)$
2	$(4,0,3,2,1)$
3	$(5,0,3,4,2,1)$
6	$(6,2,5,4,3,8,0,7,1)$.

For all other $b \ge 4$, extend $E(1,1)$ by $N = b-1$ in (29). Hence $E(b,1)$ holds for all $b \ge 1$. If $k \ge 2$ and $d \ge 3k+1$, complement $E(k-2,0)$ to get $E(k-2,k)$ and extend by $N = d-k$. When $d - k = 4k+1$, instead complement $E(k-1,1)$ and extend by $N = d-k-1 = 4k$. Thus $E(d-2,k)$ holds whenever

$$\begin{cases} d \ge 6, & k = 1, \\ d \ge 3k+1, & k \ge 2. \end{cases} \tag{30}$$

Set $k = \lceil c/2 \rceil$. To $E(d-2,k)$ graft the alpha path $(c+1, 0, c, 1, c-1, 2, \dots)$ on $c+2$ vertices, of cut $\lfloor c/2 \rfloor$. Its first label satisfies the upper identity in (18), and its second vertex, now y_d, is zero.

If both the long-tail condition and (30) fail, the remaining pairs, apart from $(1,2)$, are exactly

d	1	2	3	4	5	6	7	9	11	15
c	1	1	1, 2, 3	1, 2	1, 2, 3, 4	3	5	5, 6	7	9.

These 17 fork certificates are in `H1FiniteQ2`, and their exhaustiveness is the integer-inequality calculation in `Q2FiniteCover`. This completes the proof of Theorem 7.1.

For precision about the computer-assisted part, a fork certificate with $N = a + b + d + c$ lists labels $f_0, \dots, f_N$ in the order specified at the start of this section. Its parent function is

$$\pi(j) = \begin{cases} j-1, & 1 \le j \le a+b, \\ a, & j = a+b+1, \\ j-1, & j > a+b+1. \end{cases}$$

The finite predicate checks $f_{a+b+d} = 0$, that the f_j permute $[0, N]$, and that $|f_j - f_{\pi(j)}|$ for $1 \le j \le N$ permute $[1, N]$. The implementation supplies inverse tables v, e and checks $0 \le f_j \le N$, $v(f_j) = j$, and $e(|f_j - f_{\pi(j)}|) = j - 1$; these identities imply the two inventories. For an alpha-path certificate, the same test uses $\pi(j) = j - 1$ and also checks every edge crosses its stated cut, together with (20) or (22). For the 30 core seeds, the finite predicate is exactly the symbol and index-sum inventory and endpoint specification given in the proof of Lemma 7.2. These are finite, directly decidable predicates, and the companion file contains their witnesses and kernel-checked proofs. The construction is assembled in `Bounty.rootedFork_all`.

8. Trees with three vertices of degree three

Theorem 8.1. *Let T be a finite tree with exactly three vertices of degree three and all other vertices of degree at most two. Then T is graceful.*

Write $T_3(a, b, c; d, e; f, g)$ for the following seven-parameter tree, with all parameters positive. Its branch vertices A, M, B lie in that order on a path, with $\operatorname{dist}(A, M) = d$ and $\operatorname{dist}(M, B) = e$. Two pendant paths of lengths a, b meet A, one of length c meets M, and two of lengths f, g meet B. Thus its edge count is

$$q = a + b + c + d + e + f + g.$$

All seven paths have disjoint interiors.

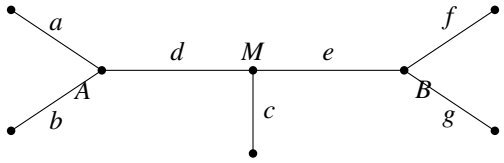


Figure 2. The tree $T_3(a, b, c; d, e; f, g)$. Every segment denotes a path of the indicated positive length.

Lemma 8.2. *Every tree in Theorem 8.1 is isomorphic to some $T_3(a, b, c; d, e; f, g)$ with positive parameters.*

Proof. The degree sum gives five leaves. Suppress all degree-two vertices, recording the positive length of each resulting edge. This produces a tree whose vertices are the five leaves and the three original branch vertices. The subgraph on its three branch vertices is connected: the path between two such vertices has no leaf in its interior, and there are no other types of vertex left. Consequently that subgraph is a three-vertex path A, M, B. Their remaining incident edges lead to two leaves at A, one at M, and two at B. Restoring the recorded subdivisions gives exactly the stated model. □

We first record how a zero can be transported along an added stem. If a q-edge tree has a graceful labeling h with $h(v) = 0$, adjoin a new leaf z at v, replace each old label by $q + 1 - h$, and label z by zero. The old differences remain $[1, q]$ and the new edge has difference $q + 1$. If h was an α-labeling of cut k, the new labeling is also α, with cut $q - k$. Iteration therefore transports zero to the end of a stem of any prescribed length, preserving the α property when it is present.

Lemma 8.3 (Peeling an outer pair)**.** *The tree $T_3(a, b, c; d, e; f, g)$ is graceful if either*

(i) $(a, b) \neq (2, 2)$ and the rooted fork $F(f, g; e, c)$ has a graceful labeling with its root zero; or

(ii) $(f, g) \neq (2, 2)$ and $F(a, b; d, c)$ has such a labeling.

Here $F(u, v; s, t)$ denotes $S(u, v, s+t)$ rooted at the vertex at distance s on its third leg, as in Theorem 7.1.

Proof. For (i), take the path consisting of the two outer arms at A. It has $a + b + 1$ vertices, and A is at position a. The prescribed-zero path theorem gives an α-labeling with $A = 0$ unless $a = b = 2$. The rest of the tree consists of $F(f, g; e, c)$, rooted at M, and the stem of length d from M to A. Transport zero along that stem. These two labeled trees meet only at A, where both labels are zero; Lemma 2.1 applies. Interchanging the two ends of the spine gives (ii). □

Lemma 8.4 (Four outer arms of length two)**.** *For all positive c, d, e, the tree $T_3(2, 2, c; d, e; 2, 2)$ is graceful.*

Proof. Suppose first that $(e, c) \notin \{(1, 1), (1, 4), (1, 5)\}$. Theorem 6.3 gives an α-labeling of the right-hand spider $S(2, 2, e + c)$ with $M = 0$. Transport zero along the d-edge stem to A, preserving the α property. The two remaining arms at A form P_5; in the order (center, first arm, second arm), the tuple

$$(0, 4, 1, 2, 3)$$

is graceful, has center zero, and has edge differences $4, 3, 2, 1$. Amalgamation at A finishes this case. The same argument from the other end works if (d, c) is outside the exceptional set.

If neither end is available, necessarily $d = e = 1$ and $c \in \{1, 4, 5\}$. The last three rows of Table 2 give complete graceful labelings of these trees. □

To specify the finite certificates without ambiguity, number the spine vertices $0, 1, \ldots, d + e$, so that $A = 0$, $M = d$, $B = d + e$. Next list the a-arm and b-arm from A, then the c-arm from M, and finally the f-arm and g-arm from B; list each arm from its attachment vertex outwards, omitting the already listed attachment vertex. Thus vertex $j > 0$ has parent

$$\pi_3(j) = \begin{cases} 0, & j = d + e + 1 \text{ or } d + e + a + 1, \\ d, & j = d + e + a + b + 1, \\ d + e, & j = d + e + a + b + c + 1 \text{ or } d + e + a + b + c + f + 1, \\ j - 1, & \text{otherwise.} \end{cases}$$

$(a, b, c; d, e; f, g)$	Label tuple
$(1, 1, 2; 1, 1; 1, 1)$	$(6, 0, 5, 3, 2, 8, 1, 7, 4)$
$(2, 2, 1; 1, 1; 2, 2)$	$(10, 0, 11, 1, 8, 5, 6, 2, 7, 4, 3, 9)$
$(2, 2, 4; 1, 1; 2, 2)$	$(9, 0, 10, 7, 12, 6, 5, 14, 1, 13, 2, 3, 11, 4, 8)$
$(2, 2, 5; 1, 1; 2, 2)$	$(10, 0, 8, 6, 9, 4, 11, 15, 1, 14, 2, 13, 7, 5, 3, 12)$

TABLE 2. The four complete-tree certificates used in the final reduction.

Each row is a permutation of $[0, q]$, and the numbers $|\ell_j - \ell_{\pi_3(j)}|$, $1 \leq j \leq q$, are a permutation of $[1, q]$. For example, the first row gives, in that order, $(6, 5, 3, 4, 8, 7, 2, 1)$.

Proof of Theorem 8.1. Apply Lemma 8.2. If both outer pairs equal $(2,2)$, use Lemma 8.4. If exactly one pair equals $(2,2)$, peel the other pair. The opposite rooted fork is covered by Theorem 7.1, since its two nondistinguished arms have length two, so it cannot be the excluded $F(1,1;1,2)$.

It remains that neither outer pair equals $(2,2)$. If $F(f,g;e,c)$ is not the excluded rooted fork, apply part (i) of Lemma 8.3. If it is excluded but $F(a,b;d,c)$ is not, apply part (ii). Finally, if both are excluded, then

$$a = b = f = g = d = e = 1, \qquad c = 2.$$

The first row of Table 2 settles this last case. This exhausts the positive parameters. □

Proof of Theorem 1.1. The degree-sum identity in the introduction reduces a five-leaf tree with at least two branch vertices to exactly one of the degree configurations $(4,3)$ and $(3,3,3)$. Theorems 1.2 and 8.1 prove gracefulness in these two cases, respectively. □

9. Lean formalization and reproducibility

The two companion files, `Source10.lean` and `Source11.lean`, independently prove the two exact catalog targets. The namespace `Math15.Graceful` contains the corresponding types, named `Target10` and `Target11`. Each quantifies over finite graphs on `Fin n`, assumes the tree property and the corresponding degree conditions, and concludes the existence of a graceful labeling. Both developments include the structural reduction from an arbitrary graph satisfying those hypotheses to its path-length model.

The final declarations have the following form in their respective files:

```
theorem target : fcTypeOfName% "Math15Catalog.source10" := by
  exact target10_of_doubleSpider_graceful
      all_double_spiders_graceful
```

and

```
theorem target : fcTypeOfName% "Math15Catalog.source11" :=
  target11_proved
```

The second proof combines `target11_of_rootedFork` with `rootedFork_all`. Thus the rooted-fork theorem is discharged inside the development, rather than retained as an additional hypothesis.

Mathematical component	Principal Lean declaration
Degree-$(4,3)$ structure	`branch43_exists_doubleSpider_iso`
All degree-$(4,3)$ models	`all_double_spiders_graceful`
Rooted three-legged spiders	`rootedFork_all`
Four outer paths of length two	`tripleSpider_twos_graceful`
Degree-$(3,3,3)$ reduction	`target11_of_positive_tripleSpider`
Complete degree-$(3,3,3)$ theorem	`target11_proved`

The finite assertions in the proof are verified using explicit lists of labels and kernel-evaluated decidable predicates. A rooted graceful certificate checks that the vertex labels are a permutation of the required interval, that the edge differences are a permutation of the positive interval, and that the distinguished label is zero. An alpha certificate additionally checks every edge against the stated cut. The symbolic recurrences, arithmetic region divisions, and structural graph arguments establish the infinite cases; they are not extrapolations from bounded tests. The companion Lean files contain the complete data for the finite checks cited in the text.

Both files passed the pinned local challenge comparator: the solution and trusted challenge built successfully, the target statements agreed, and Lean's default kernel accepted the exported proof. Only the permitted foundational axioms `propext`, `Classical.choice`, and `Quot.sound` occur in the recorded axiom checks. Independent traversals of the proof dependencies also find no catalog declaration in either closure; the catalog target is not used as a premise. Neither proof file contains `sorry`.

The checks used Lean `4.35.0-rc2` and the following catalog and mathlib revisions, respectively:

`4b69a7dca3e731dd1f28cb523b9ebd2ea32527f4`
`065356127b1dc0016f66b7283ce0ce2c4055aa55`.

Lean and mathlib are described in [2, 4]. The recorded verification took place in a local development environment; it does not assert remote platform acceptance or peer review.

The exact files used for these checks have the following SHA-256 digests. Each digest is the concatenation of its two printed lines, without punctuation or whitespace:

`Source10.lean` (9,608 lines; 479,432 bytes)
`c5ab8dc33b7b657b835e881db96c2c4a`
`0635c60747fb72049c3d3ce3eeca7761`

`Source11.lean` (16,536 lines; 995,485 bytes)
`074721668eaf52a523eeb41f77df5dbd92`
`ff0c0b8ad4cd6c4cb32b04a93c3c48`

Each file is a declaration-only submission body for its own challenge. The trusted wrapper supplies imports for `FormalConjectures.ResearchTargets.Math15` and `TaskSupport`, together with an outer `Bounty` namespace. Each complete body is therefore used separately; the two files define their own `Bounty.target` and are not concatenated into one submission. The present paper combines their mathematical conclusions as Theorem 1.1.

Acknowledgments and AI assistance

We acknowledge the authors of the prior labeling results cited above and the Lean and mathlib communities. OpenAI's Codex was used extensively in developing and checking constructions, generating and repairing Lean proofs, verifying finite labeling certificates, and drafting and typesetting this manuscript. The symbolic construction develops the approach in the earlier AI-assisted working draft supplied with this project; the constructions and structural reductions are presented here in the form used by the two verified developments. AI assistance is disclosed separately from the named authorship. Formal verification supports the specified theorems and their proof dependencies; it does not establish claims of priority.

Appendix A. Starting alpha-labelings

All tuples use order (4); index 0 is the center. The first seven rows already have $w_d = 0$. The final four rows are canonical and satisfy (6). In the tables, k denotes the α cut before normalization.

d	c_0	k	Labels in order (4)
1	2	3	5, 2, 6, 3, 4, 0, 7, 1
2	1	3	2, 5, 1, 4, 3, 7, 0, 6
3	1	3	5, 2, 7, 3, 4, 1, 8, 0, 6
4	1	4	4, 7, 2, 5, 3, 8, 1, 9, 0, 6
5	1	4	6, 3, 7, 4, 5, 1, 8, 2, 10, 0, 9
6	1	5	4, 10, 2, 6, 5, 7, 3, 8, 1, 11, 0, 9
7	1	5	7, 4, 9, 5, 6, 3, 11, 2, 8, 1, 12, 0, 10
8	1	6	1, 13, 0, 12, 2, 10, 5, 9, 3, 11, 4, 7, 6, 8
9	1	7	1, 14, 0, 13, 2, 11, 3, 12, 5, 9, 4, 10, 7, 8, 6
10	1	7	1, 15, 0, 14, 2, 12, 4, 13, 3, 10, 6, 11, 5, 8, 7, 9
11	1	8	1, 16, 0, 15, 2, 13, 5, 12, 3, 14, 4, 10, 6, 11, 8, 9, 7

Appendix B. The sixty-two residual certificates

For each row, $L = d + c$, the tuple uses order (4), and the distinguished entry is at index $4 + d$. A direct check consists of verifying that the tuple is a permutation of $[0, L + 4]$, that the differences on $Bx_1, x_1y_1, Bx_2, x_2y_2, Bw_1, w_1w_2, \ldots, w_{L-1}w_L$ are $[1, L + 4]$, and that the distinguished entry is zero. These same assertions are certified in Lean.

d	c	Labels in order (4)
1	1	6, 1, 2, 3, 5, 0, 4
1	2	7, 1, 4, 6, 2, 0, 5, 3
1	3	8, 1, 5, 3, 2, 0, 6, 4, 7
1	4	9, 1, 6, 3, 2, 0, 7, 5, 8, 4
1	5	10, 1, 7, 3, 2, 0, 8, 5, 9, 4, 6
2	1	1, 6, 3, 5, 4, 7, 0, 2
2	2	1, 7, 3, 4, 2, 8, 0, 5, 6
2	3	1, 5, 4, 3, 6, 9, 0, 7, 2, 8
2	4	1, 7, 3, 6, 4, 10, 0, 8, 9, 2, 5
3	1	5, 2, 7, 3, 4, 1, 8, 0, 6
3	2	6, 2, 8, 5, 3, 1, 9, 0, 7, 4
3	3	6, 3, 7, 4, 5, 1, 10, 0, 8, 2, 9
3	4	8, 2, 10, 6, 5, 1, 11, 0, 9, 4, 7, 3
3	5	9, 4, 8, 6, 7, 1, 12, 0, 10, 3, 5, 11, 2
4	1	2, 8, 5, 4, 3, 6, 1, 9, 0, 7
4	2	2, 9, 6, 3, 5, 7, 1, 10, 0, 8, 4
4	3	5, 10, 2, 4, 6, 8, 1, 11, 0, 9, 3, 7
4	4	5, 11, 2, 4, 7, 9, 1, 12, 0, 10, 3, 8, 6
4	5	4, 9, 6, 8, 7, 10, 1, 13, 0, 11, 3, 5, 12, 2
5	1	2, 9, 4, 5, 3, 6, 7, 1, 10, 0, 8
5	2	7, 2, 10, 5, 6, 4, 8, 1, 11, 0, 9, 3
5	3	5, 11, 2, 8, 6, 4, 9, 1, 12, 0, 10, 3, 7
5	4	2, 12, 5, 6, 9, 4, 10, 1, 13, 0, 11, 3, 8, 7
5	5	2, 13, 5, 7, 8, 4, 11, 1, 14, 0, 12, 3, 9, 6, 10
5	6	2, 14, 5, 8, 9, 4, 12, 1, 15, 0, 13, 3, 10, 7, 11, 6
6	1	4, 10, 2, 6, 5, 7, 3, 8, 1, 11, 0, 9
6	2	5, 11, 2, 7, 6, 8, 4, 9, 1, 12, 0, 10, 3

Continued on the next page

d	c	Labels (continued)
6	3	9, 2, 12, 5, 6, 7, 4, 10, 1, 13, 0, 11, 3, 8
6	4	2, 13, 5, 6, 8, 7, 4, 11, 1, 14, 0, 12, 3, 9, 10
6	5	11, 2, 14, 5, 8, 9, 4, 12, 1, 15, 0, 13, 3, 10, 6, 7
6	6	5, 15, 2, 12, 9, 10, 4, 13, 1, 16, 0, 14, 3, 11, 7, 6, 8
7	1	3, 8, 4, 6, 7, 5, 11, 2, 9, 1, 12, 0, 10
7	2	12, 2, 9, 8, 5, 7, 6, 4, 10, 1, 13, 0, 11, 3
7	3	5, 13, 2, 10, 7, 6, 8, 4, 11, 1, 14, 0, 12, 3, 9
7	4	5, 14, 2, 11, 7, 6, 9, 4, 12, 1, 15, 0, 13, 3, 10, 8
7	5	2, 15, 8, 5, 9, 12, 10, 4, 13, 1, 16, 0, 14, 3, 11, 6, 7
7	6	5, 16, 2, 13, 8, 7, 11, 4, 14, 1, 17, 0, 15, 3, 12, 6, 9, 10
7	7	2, 14, 8, 11, 10, 17, 12, 4, 15, 1, 18, 0, 16, 3, 13, 6, 9, 5, 7
8	1	3, 8, 7, 6, 4, 9, 5, 12, 2, 10, 1, 13, 0, 11
8	2	8, 5, 9, 6, 7, 13, 2, 10, 4, 11, 1, 14, 0, 12, 3
8	3	2, 14, 8, 5, 6, 11, 7, 9, 4, 12, 1, 15, 0, 13, 3, 10
8	4	12, 2, 15, 8, 9, 5, 7, 10, 4, 13, 1, 16, 0, 14, 3, 11, 6
8	5	13, 5, 9, 10, 8, 2, 16, 11, 4, 14, 1, 17, 0, 15, 3, 12, 6, 7
8	6	14, 2, 17, 5, 11, 10, 7, 12, 4, 15, 1, 18, 0, 16, 3, 13, 6, 8, 9
8	7	5, 18, 2, 15, 11, 12, 7, 13, 4, 16, 1, 19, 0, 17, 3, 14, 6, 9, 8, 10
8	8	19, 5, 9, 8, 16, 2, 7, 14, 4, 17, 1, 20, 0, 18, 3, 15, 6, 12, 11, 13, 10
9	3	9, 12, 8, 7, 6, 2, 15, 5, 10, 4, 13, 1, 16, 0, 14, 3, 11
9	4	5, 9, 8, 7, 10, 13, 2, 16, 11, 4, 14, 1, 17, 0, 15, 3, 12, 6
9	5	2, 11, 14, 8, 9, 17, 5, 7, 12, 4, 15, 1, 18, 0, 16, 3, 13, 6, 10
9	6	15, 2, 18, 5, 9, 8, 10, 7, 13, 4, 16, 1, 19, 0, 17, 3, 14, 6, 11, 12
9	7	5, 16, 8, 9, 11, 19, 2, 7, 14, 4, 17, 1, 20, 0, 18, 3, 15, 6, 12, 13, 10
9	8	5, 17, 8, 11, 10, 20, 2, 7, 15, 4, 18, 1, 21, 0, 19, 3, 16, 6, 13, 9, 12, 14
10	5	12, 8, 15, 9, 10, 2, 18, 5, 7, 13, 4, 16, 1, 19, 0, 17, 3, 14, 6, 11
10	6	16, 8, 10, 13, 9, 5, 19, 2, 7, 14, 4, 17, 1, 20, 0, 18, 3, 15, 6, 12, 11
10	7	17, 2, 20, 8, 14, 5, 9, 12, 7, 15, 4, 18, 1, 21, 0, 19, 3, 16, 6, 13, 11, 10
10	8	2, 21, 11, 18, 5, 9, 8, 13, 7, 16, 4, 19, 1, 22, 0, 20, 3, 17, 6, 14, 10, 12, 15
10	9	20, 6, 14, 7, 13, 4, 16, 5, 12, 10, 15, 11, 8, 17, 0, 23, 1, 22, 2, 21, 3, 18, 19, 9
11	7	2, 12, 10, 9, 8, 21, 5, 18, 13, 7, 16, 4, 19, 1, 22, 0, 20, 3, 17, 6, 14, 11, 15
11	8	5, 12, 11, 16, 13, 20, 6, 14, 10, 15, 9, 7, 19, 1, 23, 0, 21, 4, 17, 8, 18, 2, 22, 3
11	9	1, 22, 8, 19, 6, 20, 5, 21, 4, 16, 7, 18, 12, 2, 24, 0, 23, 3, 10, 15, 17, 9, 13, 14, 11
12	9	1, 23, 6, 21, 5, 20, 7, 19, 9, 16, 8, 22, 4, 13, 2, 25, 0, 24, 3, 18, 12, 11, 14, 10, 15, 17
12	10	5, 19, 12, 20, 10, 13, 4, 21, 1, 24, 6, 18, 7, 23, 2, 26, 0, 25, 3, 22, 9, 8, 14, 17, 15, 11, 16

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